

Jeremy Liu

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Education

University of Toronto

Bachelor of Applied Science in Computer Engineering

Sep. 2025 – Expected May 2030

Toronto, ON

Work Experience

AI Engineering Intern

Aug. 2026 – Present

Futurify

Toronto, ON

- Developed an **RFP agent** with persistent context, using PostgreSQL, pgvector, and Obsidian to self-learn company knowledge.
- Built a **21-tool MCP interface** for Claude Code and Codex, enabling authenticated RFP discovery and preparation across multi-organization workspaces with approval-gated execution.

Software Engineer Intern

Apr 2026 – July 2026

Chatforce

Toronto, ON

- Increased company **SEO/AEO visibility by 4100%** organically, and increased the number of **sign-ups by 3x**
- Spearheaded a complete **product overhaul deployed across 100% of generated games**, introducing screenshot-to-game generation, image-to-component pipelines, and a Godot-to-TypeScript engine migration that improved game quality by 3x and user experience by 5x.
- Built core generation features used in Chatforce's screenshot-to-game product, supporting **LLM routing**, asset generation, auth flows, billing webhooks, and customer-facing product **APIs**

AI/ML Research Intern

Jan. 2026 – Present

MannLab, University of Toronto

Toronto, ON

- Co-authored **2 research papers** on **EEG biofeedback** and consumer **BCI safety** by conducting wearable computing and **XR** research under Professor Steve Mann, submitted to **IEEE ISMAR** and **NeurIPS**
- Built *Cocaine Buzzkill*, an XR game in Unity with real-time **EEG biofeedback**, demoed to **200+ people**
- Identified unsafe TTA adaptation patterns by auditing **8 methods** across **9,144 EEG windows** and implementing a **3-pass activation patching diagnostic** in **PyTorch** to detect artifact-mediated failure circuits pre/post adaptation

Software Engineer

Sep. 2025 – Jan. 2026

Robotics for Space Exploration

Toronto, ON

- Implemented a **ROS2-PyQt GUI** for a robotic arm with joint monitoring, camera feeds, and 3D path-planning visualization
- Wrote unit tests using ROS and contributed to the team achieving **93.96%** on the **University Rover Challenge SAR, 20+ points** above the qualification cutoff

Projects

🐞 Paper Cuts | UC Berkeley — 1st Place Overall (\$5,000) + Best UI/UX Python, Pytorch, FastAPI, Redis, AWS Trainium, Javascript

- Made a **game engine** that converts user sketches into playable **game assets** by training an **SDXL diffusion model** on AWS Trainium and developed a cascading **CNN/VLM** recognition pipeline with **Redis HNSW vector memory**.
- Trained "Moose", a **LoRA-tuned model** that understands drawings and codes them with game mechanics in under 500 ms.

🐞 Ensemble | Hack the 6ix — Top 5 Overall

Python, QNX Raspberry Pi, Qwen LoRa

- Built an **orchestra conducting system** that synchronizes audience's phones via WebSockets, using a **QNX-based Raspberry Pi**, Arduino R4 wand, and a **TinyML model** to translate physical gestures into live wand control.
- Fine-tuned and deployed a **Qwen LoRA** model on a **Fireworks H200** using SFT and GRPO over classical music theory, generating musical bars in real time with 453 ms latency.

🐞 Aucctopus | SOON 2026 Winner

TypeScript, Neo4j, OpenRouter, Composio

- Reduces iteration time on consumer products for companies by **90%**, using swarm intelligence models to evaluate **3D models**, **UGC videos**, and **AR previews** to predict virality. Optimized costs and latency for **10+ APIs**.

🐞 Straw | 50+ on waitlist

TypeScript, Supabase, BullMQ, Redis, Docker

- Task outsourcing platform where **Hermes/Openclaws** communicate and compete against one another using **APIs** and **MCPs**

Technical Skills

Languages: Python, C, JavaScript, TypeScript, MATLAB, SQL

Frameworks & Libraries: React, Next.js, Node.js, FastAPI, Tailwind, ROS2, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, OpenCV

Tools & Databases: Git, Docker, Linux, AWS, Unity, PostgreSQL, Redis, Neo4j, Supabase, Postman, Figma, Claude